

Top 25 MySQL Interview Questions

1. Explain different types of JOINS with business examples.

Competency Assessed

- SQL Fundamentals
- Relational Database Understanding

Interview-Ready Sample Answer

SQL JOINS are used to combine data from multiple related tables.

- **INNER JOIN** returns only matching records. For example, showing customers who have placed orders.
- **LEFT JOIN** returns all records from the left table, including customers who have never placed an order.
- **RIGHT JOIN** returns all records from the right table and matching records from the left.
- **FULL OUTER JOIN** returns all records from both tables, although MySQL simulates this using UNION.
- **SELF JOIN** is useful when comparing records within the same table, such as employees and their managers.

I choose the JOIN type based on the business question rather than simply retrieving data.

SQL Concepts

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- SELF JOIN
- UNION

Interview Tip

Always explain with a real business example instead of only defining the JOIN.

Example:

-- INNER JOIN: customers who have placed orders

```
SELECT c.customer_id, c.name, o.order_id
```

```
FROM customers c
```

```
INNER JOIN orders o ON c.customer_id = o.customer_id;
```

```
-- LEFT JOIN: all customers, including those with no orders
```

```
SELECT c.customer_id, c.name, o.order_id
```

```
FROM customers c
```

```
LEFT JOIN orders o ON c.customer_id = o.customer_id;
```

```
-- MySQL has no native FULL OUTER JOIN — simulate with UNION
```

```
SELECT c.customer_id, c.name, o.order_id
```

```
FROM customers c
```

```
LEFT JOIN orders o ON c.customer_id = o.customer_id
```

```
UNION
```

```
SELECT c.customer_id, c.name, o.order_id
```

```
FROM customers c
```

```
RIGHT JOIN orders o ON c.customer_id = o.customer_id;
```

```
-- SELF JOIN: employees and their managers
```

```
SELECT e.name AS employee, m.name AS manager
```

```
FROM employees e
```

```
LEFT JOIN employees m ON e.manager_id = m.employee_id;
```

MySQL doesn't support **FULL OUTER JOIN** at all — the UNION is a workaround, not a stylistic choice. Interviewers sometimes ask "why UNION and not UNION ALL here?" — answer: UNION removes duplicate rows that appear in both the LEFT and RIGHT results (the matching rows), UNION ALL would double them.

2. How would you investigate declining sales using SQL?

Competency Assessed

Business Analysis

Sample Answer

I would first clarify the business definition of declining sales. Then I would compare current sales with previous months or quarters. I would analyse revenue by product, category, region, customer segment, salesperson and channel. I would also examine order volume, average order value, discounts, cancellations and returns. Finally, I would validate whether external factors such as seasonality or promotions contributed to the decline before presenting recommendations.

SQL Concepts

- GROUP BY
- SUM()
- Date Functions
- Window Functions

Interview Tip

Focus on your analytical approach before discussing SQL syntax.

3. How would you identify duplicate customer records?

Competency Assessed

Data Quality

Sample Answer

I would first determine what defines a duplicate record, such as duplicate email addresses, phone numbers or combinations of customer attributes. Then I would use GROUP BY with COUNT() to identify duplicates. After validating with business rules, I would decide whether to merge, remove or retain the records based on the business requirement.

SQL Concepts

- GROUP BY
- COUNT()
- HAVING

- ROW_NUMBER()

Interview Tip

Don't assume every duplicate should be deleted.

Example:

-- Find duplicates by email

SELECT email, COUNT(*) AS occurrences

FROM customers

GROUP BY email

HAVING COUNT(*) > 1;

-- Flag duplicates while keeping row-level detail (to decide which to keep)

SELECT customer_id, email, created_at,

ROW_NUMBER() OVER (PARTITION BY email ORDER BY created_at) AS rn

FROM customers;

-- rn = 1 is typically the "original" record to keep; rn > 1 are candidates for merge/removal

4. A JOIN returns significantly more rows than expected. How would you investigate?

Competency Assessed

SQL Debugging

Sample Answer

I would first verify the expected number of records. Then I would inspect the JOIN conditions and primary-foreign key relationships. I would check for one-to-many relationships that naturally increase row counts and isolate each JOIN individually to identify where duplication occurs. Only after understanding the cause would I consider aggregation or DISTINCT.

SQL Concepts

- JOIN
- DISTINCT
- GROUP BY

5. How would you optimize a slow SQL query?

Competency Assessed

Performance Optimisation

Sample Answer

I would examine the execution plan using EXPLAIN to identify bottlenecks. I would ensure indexes exist on frequently filtered and joined columns, avoid SELECT *, apply filters as early as possible, and simplify complex joins or subqueries where appropriate. After optimisation, I would compare execution times and validate that the business results remain unchanged.

SQL Concepts

- EXPLAIN
- Indexes
- Query Optimisation

6. What checks do you perform before analysing data?

Sample Answer

Before analysing data, I verify record counts, identify missing values, check for duplicates, validate primary and foreign key relationships, review data types, inspect date ranges, and confirm that the data aligns with business definitions. This helps ensure the analysis is based on reliable data.

7. How would you validate SQL results before sharing them?

Sample Answer

I compare totals with trusted reports, manually verify sample records, reconcile row counts, validate joins and aggregations, review business logic with stakeholders, and test edge cases. I only share results once both technical and business validations are complete.

8. How would you analyse customer purchasing behaviour?

Sample Answer

I would analyse purchase frequency, recency, average order value, lifetime value, repeat purchases, product preferences and seasonal buying patterns. This helps identify loyal customers, churn risks and cross-selling opportunities.

9. How would you calculate business KPIs using SQL?

Sample Answer

I would calculate KPIs such as total revenue, profit, average order value, customer acquisition, customer retention, repeat purchase rate, conversion rate and monthly growth using SQL aggregations and date functions before visualising them in reporting tools.

10. Explain Primary Key and Foreign Key.

Sample Answer

A Primary Key uniquely identifies each record in a table and prevents duplicates or NULL values. A Foreign Key links one table to another, ensuring referential integrity by allowing only valid relationships between related tables.

11. When would you use Common Table Expressions (CTEs) instead of subqueries?

Competency Assessed

- Advanced SQL
- Query Structuring

Sample Answer

I use Common Table Expressions (CTEs) when I want to make complex SQL queries more readable, modular, and easier to maintain. CTEs allow me to break a large query into logical steps, making it easier to debug and modify. They are especially useful when the same intermediate result needs to be referenced multiple times or when working with recursive queries. While subqueries are suitable for simple calculations, CTEs improve readability and collaboration, particularly in analytical queries involving multiple transformations. However, I also consider query performance, as in some scenarios a subquery or temporary table may perform better depending on the database engine and execution plan.

SQL Concepts

- CTE
- WITH clause
- Query readability
- Recursive CTE

Interview Tip

Emphasise readability and maintainability rather than saying CTEs are always faster.

12. Explain Window Functions with business examples.

Competency Assessed

- Advanced SQL
- Business Reporting

Sample Answer

Window functions perform calculations across a set of related rows without collapsing them into a single result, unlike GROUP BY. They are useful for business reporting where row-level details need to be retained. For example, I can rank salespeople by revenue, calculate running totals, compare a product's sales with the previous month using LAG(), or calculate the average sales within each region. Window functions simplify analytical queries and reduce the need for complex self-joins or subqueries, making reports more efficient and easier to understand.

SQL Concepts

- ROW_NUMBER()
- RANK()
- DENSE_RANK()
- LAG()
- LEAD()
- OVER()

Interview Tip

Always explain that window functions retain individual rows while performing calculations.

13. How would you identify the second highest-selling product?

Competency Assessed

- SQL Problem Solving
- Ranking Functions

Sample Answer

I would first calculate the total sales for each product using GROUP BY and SUM(). Then, I would apply a ranking function such as DENSE_RANK() or RANK() in a CTE or subquery to rank products based on total sales. Finally, I would retrieve the product with a rank of two. I prefer using ranking functions because they correctly handle ties, making the solution more reliable than using nested MAX() functions.

SQL Concepts

- GROUP BY
- SUM()
- DENSE_RANK()
- CTE

Interview Tip

Mention how ties should be handled—it shows attention to business requirements.

Example:

```
WITH product_sales AS (  
    SELECT product_id, SUM(sale_amount) AS total_sales  
    FROM sales  
    GROUP BY product_id  
)  
  
ranked AS (  
    SELECT product_id, total_sales,  
           DENSE_RANK() OVER (ORDER BY total_sales DESC) AS sales_rank  
    FROM product_sales  
)  
  
SELECT product_id, total_sales  
FROM ranked
```

WHERE sales_rank = 2;

14. How would you prepare a monthly management report?

Competency Assessed

- Business Reporting
- SQL Analytics

Sample Answer

I would first understand the KPIs required by management, such as total revenue, profit, order count, customer growth, and regional performance. Using SQL, I would aggregate the data monthly, validate the results against existing reports, and ensure consistency in business definitions. I would also calculate month-over-month growth, identify top-performing products and regions, and highlight any significant trends or anomalies. Finally, I would present the cleaned dataset to reporting tools like Power BI or Excel, ensuring the report is accurate, easy to understand, and actionable.

SQL Concepts

- GROUP BY
- SUM()
- COUNT()
- Date Functions

Interview Tip

Focus on business insights, not just report generation.

15. How would you identify top-performing customers?

Competency Assessed

- Customer Analytics
- Business Intelligence

Sample Answer

I would define the criteria for "top-performing" customers based on the business objective. This could include highest revenue, highest profit, most frequent

purchases, highest customer lifetime value, or repeat purchase rate. I would aggregate customer transactions, rank customers using window functions, and analyse purchasing behaviour over time. I would also consider profitability and customer retention instead of relying solely on revenue, as high sales do not always translate into high business value.

SQL Concepts

- SUM()
- COUNT()
- GROUP BY
- Window Functions

Interview Tip

Clarify that "top-performing" depends on business goals.

16. How would you detect missing or invalid business data?

Competency Assessed

- Data Quality
- Validation

Sample Answer

I would identify NULL values, blank fields, duplicate records, invalid dates, incorrect formats, and values outside expected business ranges. I would compare related tables to identify orphan records and verify referential integrity using primary and foreign keys. I would also review record counts, mandatory fields, and business rules before analysis. Any issues identified would be documented and discussed with stakeholders before making assumptions or corrections.

SQL Concepts

- IS NULL
- COUNT()
- CASE
- JOIN

Interview Tip

Mention business validation in addition to technical validation.

17. How would you reconcile differences between two reports?

Competency Assessed

- Analytical Thinking
- Data Validation

Sample Answer

I would first confirm that both reports use the same data source, reporting period, filters, and business definitions. Then I would compare row counts, aggregations, joins, and calculations to identify where the differences arise. I would reconcile data step by step, validate sample records, and discuss any differences in business logic with stakeholders. My objective would be to identify the root cause rather than simply matching the numbers.

SQL Concepts

- JOIN
- EXCEPT (or alternative approaches)
- Aggregations

Interview Tip

Show a structured investigation rather than guessing the cause.

18. How would you explain SQL analysis to non-technical stakeholders?

Competency Assessed

- Communication
- Stakeholder Management

Sample Answer

I would avoid technical SQL terminology and focus on business outcomes. Instead of explaining joins or window functions, I would describe what the analysis reveals, why it matters, and how it supports decision-making. I would use simple language, visualisations, and key metrics to communicate insights. If stakeholders have questions, I would explain the methodology at a high level while ensuring the conclusions remain transparent and easy to understand.

SQL Concepts

- Business Communication

Interview Tip

Interviewers value communication skills as much as technical skills.

19. How would you troubleshoot a SQL query returning zero records?

Competency Assessed

- SQL Debugging

Sample Answer

I would first verify that the source tables contain data. Then I would review WHERE conditions, JOIN criteria, date filters, and parameter values to ensure they are not excluding valid records. I would execute the query step by step, testing each condition individually to identify where records are being filtered out. I would also verify data types and confirm that the business assumptions match the available data before concluding that no records exist.

SQL Concepts

- WHERE
- JOIN
- Debugging

Interview Tip

Explain a systematic debugging process instead of immediately changing the query.

Example:

-- Step 1: confirm the source table actually has data

```
SELECT COUNT(*) FROM orders;
```

-- Step 2: test the WHERE clause alone, no joins

```
SELECT COUNT(*) FROM orders WHERE order_date >= '2026-01-01';
```

-- Step 3: test each JOIN condition in isolation

```
SELECT COUNT(*) FROM orders o
```

```
INNER JOIN customers c ON o.customer_id = c.customer_id;
```

-- Step 4: check for data type / format mismatches (a very common silent cause)

```
SELECT DISTINCT order_date, CAST(order_date AS CHAR) FROM orders LIMIT 5;
```

20. How would you optimize queries on millions of records?

Competency Assessed

- Performance Optimisation

Sample Answer

When working with large datasets, I minimise the amount of data processed by filtering records as early as possible and selecting only the required columns instead of using `SELECT *`. I ensure appropriate indexes exist on frequently filtered and joined columns, avoid unnecessary subqueries, and review execution plans using `EXPLAIN`. Where appropriate, I use partitioning, aggregation, or precomputed summary tables to reduce processing time. I always measure performance before and after optimisation to confirm that improvements do not affect data accuracy.

SQL Concepts

- `EXPLAIN`
- Indexes
- Partitioning

Interview Tip

Always mention validating performance improvements.

Example:

-- Check the execution plan first, always

```
EXPLAIN SELECT customer_id, SUM(order_total)
```

```
FROM orders
```

```
WHERE order_date >= '2026-01-01'
```

```
GROUP BY customer_id;
```

-- Composite index example: filters and joins should match index column order

```
CREATE INDEX idx_orders_date_customer ON orders(order_date, customer_id);
```

-- Precomputed summary table for a recurring report (instead of re-aggregating millions of rows every run)

```
CREATE TABLE monthly_customer_sales AS
```

```
SELECT customer_id, DATE_FORMAT(order_date, '%Y-%m-01') AS month,
```

```
       SUM(order_total) AS total_sales
```

```
FROM orders
```

```
GROUP BY customer_id, DATE_FORMAT(order_date, '%Y-%m-01');
```

21. How have you used ChatGPT or GitHub Copilot while writing SQL?

Competency Assessed

- AI Productivity

Sample Answer

I use AI tools such as ChatGPT or GitHub Copilot to accelerate query development, understand unfamiliar SQL syntax, generate sample queries, explain complex window functions, and troubleshoot errors. They are also helpful for documenting SQL logic and suggesting alternative approaches. However, I always validate AI-generated code by reviewing the logic, testing it in a development environment, and verifying the results against business expectations before using it in production.

SQL Concepts

- AI-assisted SQL Development

Interview Tip

Emphasise AI as a productivity tool, not a replacement for SQL knowledge.

22. How would you validate AI-generated SQL?

Competency Assessed

- AI Governance
- SQL Validation

Sample Answer

I would review the query for correct table names, joins, filters, aggregations, and business logic before execution. I would test the query in a development environment, compare the output with trusted reports or manually calculated results, and review the execution plan if performance is important. Only after confirming both technical accuracy and business correctness would I consider using the query in production.

SQL Concepts

- Query Validation
- Testing
- Execution Plan

Interview Tip

Use the phrase **"Trust AI, but always verify."**

23. Can AI replace SQL developers?

Competency Assessed

- Critical Thinking
- AI Awareness

Sample Answer

AI can significantly improve productivity by generating SQL queries, explaining syntax, suggesting optimisations, and automating repetitive tasks. However, it cannot replace SQL developers or analysts because it lacks business context, domain knowledge, and accountability. Professionals are still responsible for understanding requirements, validating results, ensuring data quality, optimising performance, and making business decisions. AI works best as a collaborative assistant rather than a replacement for skilled analysts.

SQL Concepts

- AI Productivity

Interview Tip

Present AI as an enabler, not a substitute.

24. AI-generated SQL gives incorrect business numbers. What will you do?

Competency Assessed

- Problem Solving
- Validation

Sample Answer

I would first compare the AI-generated results with trusted reports or manually calculated values to confirm the discrepancy. Then I would review joins, filters, aggregation logic, date ranges, and business assumptions to identify the cause. I would execute smaller portions of the query to isolate the issue and consult stakeholders if business definitions are unclear. Once the root cause is identified, I would correct the query, retest it, and document the findings to avoid similar issues in the future.

SQL Concepts

- Query Validation
- Debugging
- Business Rules

Interview Tip

Never assume AI-generated results are correct simply because the query executes successfully.

25. Describe an end-to-end SQL project you have worked on.

Competency Assessed

- Project Experience
- Business Problem Solving

Sample Answer

In one of my projects, I analysed sales data to create a business performance dashboard. I began by understanding stakeholder requirements and identifying the necessary data sources. I used SQL to clean the data, join multiple tables, calculate KPIs such as revenue, profit, and customer retention, and validate the results against existing reports. The processed data was then integrated into Power BI to create interactive dashboards. Throughout the project, I optimised queries for performance, documented the SQL logic, and presented actionable insights to stakeholders, enabling them to monitor business performance and make informed decisions.

SQL Concepts

- JOIN
- GROUP BY
- Window Functions
- CTE
- Data Validation
- Performance Optimisation

Interview Tip

Structure your answer as **Business Problem** → **SQL Solution** → **Business Impact**, as interviewers value the outcome as much as the technical implementation.